From Automation to Scaffolding: Rethinking Generative UI for Design Upskilling

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1 The Provocation: Democratization of UI Creation vs. Design Thinking

The emergence of Generative User Interface (GenUI) models has initiated a paradigm shift in User Interface(UI) design, enabling anyone to transform ideas into professional-looking UI prototypes instantly [1, 5, 6]. This "democratization of UI generation" empowers non-experts to create high-fidelity prototypes rapidly that was previously the domain of skilled practitioners. However, as we envision the future of HCI practice, we must ask: **Does the democratization of *output* equate to the democratization of *skill*?**

Past literature warns that the efficiency of AI often comes at the cost of steep cognitive reduction [3, 4, 8, 9]. Design is no exception to this. Traditional UI design learning relies on a scaffolded journey: moving deliberately from user journey maps and user flows to information architecture (IA) and wireframes. These intermediate steps are not only for the final products, but also for the students to learn to translate abstract user needs into concrete interface logic. However, GenUI allows learners to leapfrog directly to high-fidelity outcomes, threatening to collapse these critical cognitive processes into a single instantaneous inference. Critically, this risks creating an "illusion of competence" where learners mistake prompt proficiency for design mastery. If the next generation of designers is trained with heavy reliance on GenUI, **how can we ensure we are not educating a generation of "Hollow Designers" who possess the tools to create but lack the cognition to construct?**

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Through a formative study with design novices, we reveal that without intervention, GenUI risks turning learners into passive curators of automated output rather than active architects of user experience. We argue that the current paradigm of GenUI, optimized for one-click outcomes, inadvertently incentivizes users to bypass the essential work of design thinking. In this position paper, we propose that for GenUI to ensure truly human-centric experiences where human capabilities are nurtured rather than eroded or replaced, we need to rethink GenUI to evolve from an execution tool into a scaffold that prioritizes upskilling and metacognition.

2 Empirical Evidence

Our recent study provides empirical evidence for this concern. We interviewed design professionals (N=7) to understand expert practice, followed by a longitudinal task-based study with novice learners (N=7) who interact with and without GenUI during their design. We observed that while GenUI tools (e.g., FigmaMake [1]) lowered the visualization and prototyping barrier, current workflows fail to support design upskilling and, in some cases, entrench shallow design habits:

Bypassing Structure for Visuals: We observed a strong tendency among novices to prioritize surface visual details (colors, fonts) over UX architecture (e.g., user flow, IA). Three novice participants prompted GenUI solely to change visuals, using the tool to "paint" screens rather than "architect" flows. This suggests that without scaffolding, GenUI reinforces a visual-focused mental model that neglects the underlying logic prioritized by professionals.

Intuition-Based Decisions without Rationale: When asked to justify design decisions, novices relied on intuition (e.g., *"I felt it was natural"*) rather than design principles. GenUI exacerbated this by generating visuals without much explanation, leaving users unable to articulate *why* a design worked or failed. Consequently, decisions to accept or reject GenUI outputs were based on subjective preference rather than informed rationale.

The "Template" Trap: Critically, code-based GenUI tools tended to produce generic, "cookie-cutter" designs lacking in creativity, potentially due to their usage of existing code libraries for rendering UI. We observed that GenUI risks diminishing human creativity by homogenizing participants' original designs into standardized template-like ones.

[1] https://www.figma.com/make/

Stagnant Exploration and Iteration: Professional design practice emphasizes exploring multiple solutions and iteration. However, when novices collaborated with GenUI, no participants leveraged GenUI's speed to explore parallel alternatives [2]. We are concerned that the immediate provision of high-fidelity outputs by GenUI may erode users' motivation for exploration and iteration.

3 Envisioning Future Practice: Rethinking GenUI From Automation To Upskilling

To address the risk of the "hollow designer," we propose that the HCI community should **reimagine GenUI as an upskilling scaffolding mechanism** for design learners, not just an efficiency engine. We envision a GenUI shift for design learners from standard one-click generation, which can bypass critical thinking, to process-aware scaffolding that actively cultivates design mastery. Future learner-centric tools should decompose one-click generation into pedagogical stages, such as enforcing a workflow that requires users to define and verify the user flow and IA before rendering any visuals. By constraining the leap directly to visuals, we force novices to engage with the structural logic that underpins effective design.

Moreover, to counter the "template" trap and stagnant iteration, learner-centric GenUI should discourage the fixation on a single design output. Instead, it should prompt learners to explore multiple divergent iterations in parallel [2] through methods like visualizing design space maps [7, 10] rather than directly giving one solution output. This nudges novices to compare trade-offs, fostering an exploratory mindset that values iteration over immediate completion.

Finally, future learner-centric GenUI should transition from an executor to an instructor. AI interventions must move beyond merely describing visual attributes (e.g., "generated 8px padding") to explaining the design rationale and pedagogical purpose behind every design output. Thus, we can help learners engage with foundational design thinking and internalize the *why* behind the UI.

As GenUI reshapes workflows, we face a choice: allow these tools to become automators that hollow out skills, or redesign them as scaffolds that augment human intelligence rather than atrophy it. By redesigning GenUI to prioritize upskilling, we can ensure that the democratization of UI generation leads to the true democratization of design mastery.

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